ADAGIO GROUP

Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Re: Technical review of Paper 0/1 - Physics of the Instant

Benjamin D. Summers <bsummers@adagiollc.com> Mon, Sep 14, 2026 at 12:52 PM
To: Ezequiel Boero <ezequiel.boero@adagioinstitute.org>

Hi Ezequiel,

Thank you for the report. The revised manuscript has been uploaded to SSRN (https://dx.doi.org/10.2139/ssrn.6922380). Below are my responses to each finding and question, in manuscript order, starting with the one that explains most of the others. The Section 3 scoping, the clock-bound arithmetic, and the renormalization corrections you confirmed are unchanged.

Two items about the report itself. Question 5 refers to "the two strengthenings of Finding ??" and to a sampling floor, and Finding 2 and the technical conclusion refer to a counterexample "recorded in the verification section." Neither the finding nor the section is in the report I received. If you send them, I will answer Question 5 and check the counterexample against Lemma 1 in a follow-up. Also as a heads up, the report is dated 4 September and examines the version of 7 September.

The one that explains most of the others is Finding 6. Dynamical completeness was never defined, and without a definition the abstract, Section 6, and Section 7 ran the Markov property, Chapman-Kolmogorov composition, and completeness together as one condition. They are not one condition, and the paper now says so where the notions first appear. Section 1 carries Definition 1: in the deterministic setting, a well-posed initial-value problem from data on a single slice; in the stochastic setting, the Markov property with respect to the natural filtration. The paragraph after it states the hierarchy: the Markov property implies that the two-time kernels compose, the converse fails by Feller's 1959 example, so composition is a necessary condition for completeness in the stochastic sense, its failure refutes completeness, and its holding does not establish it. The same paragraph says which sense each section concerns: Sections 2 to 4 bear on both, Section 5 on the quantum state in each formulation, Section 6 on the stochastic sense on a specified observed state space. This answers Question 1. Question 2 is answered by the same hierarchy: a Feller-type process shows that composition is necessary and not sufficient; every inference in Section 6 runs in the necessary direction; and the test in Section 6 is now stated as one-sided, since such a process passes it while being non-Markov. Composition is described throughout as the observable through which the Markov property is tested, and nowhere as a description that presupposes completeness.

The second change that runs through the paper is the window claim, Finding 2. The unconditional sentence in Section 2 was false, for the reason you gave: if the observation is a non-lumpable function of the state, the defect converges to that of the lumped family and no refinement removes it. Section 2 now names the operation in force. A temporal window is a lumping of the path-segment process and not of the state, and the path-segment process of a Markov process is itself Markov, so Burke-Rosenblatt and Kemeny-Snell are cited for what they establish, the existence of a defect, and for nothing about refinement. The refinement behavior is Lemma 1 in Section 6, which answers Questions 3 and 4. For an irreducible finite-state chain started in its stationary law, with maximal jump rate $q_$, every row of the composition defect of the window average has total variation at most $C\, q_\, \delta$ for $\delta \le \min\{(t - t_1)/2, 1/(2q_)\}$, with $C = 2 + 8/c$, where c is the smaller of the minimal stationary probability and the minimal transition probability over lags between $(t - t_1)/2$ and $t - t_1$. The proof uses the Markov property at t_1 and at t and the bound $q_\, \delta$ on the probability of a jump inside the window; the constant is explicit and the rate is linear. The second half of the lemma states the converse case: for the window average of a non-lumpable function of the state, the kernels converge to those of the lumped process and the defect converges to the lumped family's, which is generically positive and independent of δ. I verified the linear rate in exact rational arithmetic on a two-state chain before writing it: every row of the defect, including the row conditioned on the rare intermediate value, scales linearly with the width.

1. Abstract (Findings 4, 5, and 6). Accepted. "Physical predictions are defined at finite resolution" is replaced by the validity-scale statement. The algebraic clause now says that the formulation assigns observables to open regions, fixes the dynamics from any neighborhood of a Cauchy surface, and assigns no algebra to the surface itself. The closing sentences state the definition, the hierarchy, and the one-sided test.
2. Section 2, the apparatus (Finding 1, and the remark in the overall assessment on the state as a primitive notion). Accepted in part. You are right that the conclusion follows from the meaning of sufficiency, and the section now says so in its first paragraph. The interval average is defined explicitly in equation (1), with the convergence condition stated, which is the ground the "element of the state space" sentence was missing. The section is tightened, not cut. I kept it because it is where the two stipulations, continuity in time and dynamical sufficiency, are located, and the paper's claim is that those two are stipulations. If you think it still outweighs its yield, tell me where to cut.
3. Section 3 (Finding 3 and Question 6). Accepted. The inference from the spatial bounds to the temporal one now runs through the position-momentum relation, as you supplied it: a signal that resolves δt is localized in time

to δt and in space to c δt, localization to c δt requires momentum $\hbar/(c\,\delta t)$ and hence energy $\hbar/\delta t$, and the text says that this route does not invoke the time-energy relation, so the disclaimer two paragraphs later is consistent with it. The scoping paragraph, the Mead-Garay and 't Hooft paragraphs, and the Salecker-Wigner derivation are unchanged; the bound is written $\delta t \gtrsim (T\, t_P^2)^{1/3}$, in units of time throughout, as you noted it should be.

4. Section 4, scales (Finding 4 and Question 8). Accepted. The section now separates the regulator, the renormalization scale μ, the validity scale Λ of an effective description, and experimental resolution, and states that exact observables are independent of μ, that being the content of the renormalization-group equation. The running of α_s is retained as confirmation of the renormalization-group structure and is no longer offered as a limit on resolution; what is measured is a scheme-dependent parameter at a chosen scale. The Wilsonian paragraph is rewritten in terms of the cutoff Λ: the description at Λ is effective, its parameters flow as Λ is lowered, and a continuum limit requires an ultraviolet fixed point of the flow. Question 8: the argument requires Λ and the coincident-point singularity; μ is not used; the regulator is removed from renormalized quantities; experimental resolution belongs to Section 3, and the phrase "finite resolution" no longer appears in Section 4.
5. Section 4, algebraic formulation (Finding 5, Question 7, and the Wightman attribution). Accepted. The time-slice axiom is now stated, with Haag-Schroer 1962 for primitive causality and Chilian-Fredenhagen 2009 for its proof in perturbatively renormalized scalar theories, and the reading you described is the one the paper gives: the dynamics is determined by a slab of nonzero thickness, however small, which is an initial-value structure; the axioms assign no algebra to the sharp surface; and it is the slab that carries the determination. That is the paper's thesis inside a rigorous formulation, and the section says so. Question 7, second half: the singularity of point-localized fields does not bear on the completeness of Cauchy data directly, and the paper no longer suggests it does; the bearing on initial-value structure comes through the time-slice axiom, and the point-localization results are confined to the localization of observables, with the explicit statement that neither bears on the composition of two-time kernels. The Wightman theorem is now stated: a field defined as an operator-valued function on spacetime, with the vacuum in its domain, translation covariant and microcausal, in a theory satisfying the spectrum condition with a unique vacuum, satisfies $\varphi(x)\Omega = c\Omega$ for every x, so a field with nontrivial vacuum correlations exists only as an operator-valued distribution. Those are the hypotheses and the conclusion; the phrase "lead to contradictions under natural assumptions" is gone.
6. Section 5 (Section 3.4 of the report). Accepted. The section now says there is no operationally privileged instant at which the measurement happens, and adds that this is not the claim that no theoretical jump time can exist, which the discrete collapse models cited in Section 2 posit and which the arguments do not exclude. The conclusions drawn from the stochastic-quantum correspondence are marked as conclusions inside that framework, not consequences of standard quantum theory.
7. Section 6, the prediction and the criterion (Section 3.5 of the report and the second positioning item). Accepted. The prediction is now the lemma, with the hypothesis stated and the test one-sided: a persistent defect refutes the memoryless description on the observed state space; a vanishing defect is consistent with it and does not establish it. Your caution that memory can arise other than from finite resolution is the lemma's second half. The criterion separating classical from quantum memory is now stated rather than deferred: consistency under non-invasive intermediate observation. A classical process observed non-invasively satisfies the Kolmogorov consistency conditions, its two-time statistics being the marginal of its three-time statistics over the unobserved intermediate value; a quantum process on its configuration space generically violates them, an intermediate measurement whose outcome is discarded altering the later statistics, which is what the process-tensor framework isolates (Milz and Modi). The criterion inherits the non-invasiveness assumption of Leggett-Garg tests, and the text says so.
8. Section 7 (Section 3.6 of the report, Questions 9 and 10, and the first positioning item). Accepted. "Without leaving the domain of confirmed physics" is replaced by a sentence consistent with Section 3: the arguments invoke no physics beyond the established theories, and the Planck-scale bounds extrapolate two of them beyond their tested domains. The list of descriptions that presuppose completeness is corrected per Finding 6. Question 10 is answered in the last paragraph: the foundational claim is a claim about warrant and is not itself an empirical prediction; what is empirically at stake is the Planck-scale bound, which a measurement resolving time below t_P would refute, and the refinement behavior of Lemma 1, falsifiable in the direction stated there. The positioning gap is closed with one paragraph: Butterfield's argument against pointillisme (2006), following Arntzenius (2000), is the nearest neighbor; it concerns the intrinsicness of a quantity, instantaneous velocity, and the present claim concerns the dynamical sufficiency of the state and rests on the Planck-scale bounds and field-theoretic localization rather than on the analysis of velocity. The two are compatible.
9. There is one recommendation I did not adopt in full, the shortening of Section 2, noted above.
10. On originality. Accepted for the three items you credit as the paper's own and for the priority finding. Not accepted for the sentence that grades them. Section 7 says the novelty is in the joint use of standard results and not in the results. That locates the originality; it does not grade it, and "low as new mathematics or new physics" is your grade, not the paper's. It is also the wrong yardstick, and your own summary says why: the paper is conceptual and inferential, and its object is a presupposition carried by every initial-value formulation in physics. A paper whose contribution is to show that a universal presupposition is unwarranted cannot score as new physics if new physics means new results. By that standard EPR contributed none either. The axis that applies is the foundational one, and there your grade of moderate sits next to your finding that no prior work assembles these lines or proposes the criterion. I want to be direct: I am not claiming the paper contains new physics. The revision adds one result, Lemma 1, new mathematics of modest scope with an explicit rate, and nothing else. What I am asking is what a high grade would have required of a paper of this kind. If it is a result the thesis licenses and the presupposition forbids, say so. If it is something else, say what.

Bibliography: added are Feller 1959, Haag-Schroer 1962, Chilian-Fredenhagen 2009, Leggett-Garg 1985, Butterfield 2006, and Arntzenius 2000. Nothing is removed. The Wightman entry is unchanged and the attribution is now the theorem. The paper cites no companion paper.

Let me know your thoughts.

Ben

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On Fri, Sep 11, 2026 at 7:30 PM Ezequiel Boero <ezequiel.boero@adagioinstitute.org> wrote:

> Ben,
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> Please I'm sending you attached the technical review of paper 3 of the submitted portfolio:
> "The Physics of the Instant: Analysis from Limits, Relativity, and Renormalization"
>
> Ezequiel